PROJECT REPORT: VIT-BHOPAL-AI-CONCIERGE

Project Name: VIT-Bhopal-AI-Concierge

Domain: Artificial Intelligence / Educational Technology

Platform: Web-based (Integrated Agent)

Live Demo: View Live Project

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1. Executive Summary

The **VIT-Bhopal-AI-Concierge** is a next-generation communication interface designed for VIT Bhopal University. It addresses the challenge of information discovery on complex university portals by providing a dual-mode (Text & Voice) AI agent. By utilizing Retrieval-Augmented Generation (RAG), the system ensures that responses are grounded in real-time, official university data.

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2. Key Features & Functionalities

2.1 Dual-Interface Interaction

- **AI Chatbot:** A structured text interface for detailed queries.
- **AI Voice Agent:** A real-time vocal assistant for hands-free conversational engagement.

2.2 Knowledge Domains

The agent is specialized in the following university sectors:

- **Academics:** Branch specializations, FFCS, and faculty details.
- **Admissions:** Fees, scholarship eligibility, and important dates.
- **Placements:** Salary packages and recruiter lists.
- **Campus Life:** Hostel ("Lion's Den") and cultural fests like AdVITya.

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3. Project Visuals

3.1 AI Chat Bot Interface

The chatbot is docked in the bottom-right corner and provides a clean, conversational UI for navigating university data.

Users can interact with a specialized assistant that understands specific institutional nuances.

3.2 AI Voice Agent Mode

The Voice mode allows for a hands-free experience, perfect for mobile users or quick verbal inquiries.

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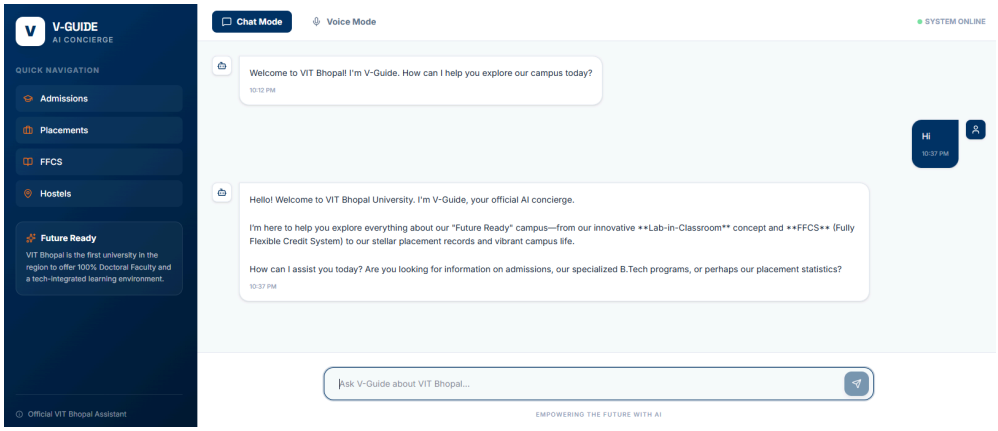
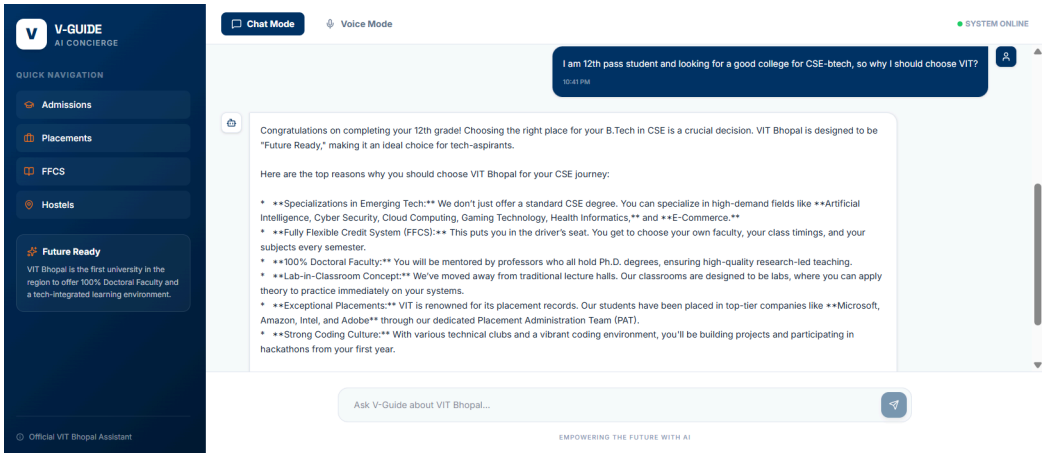
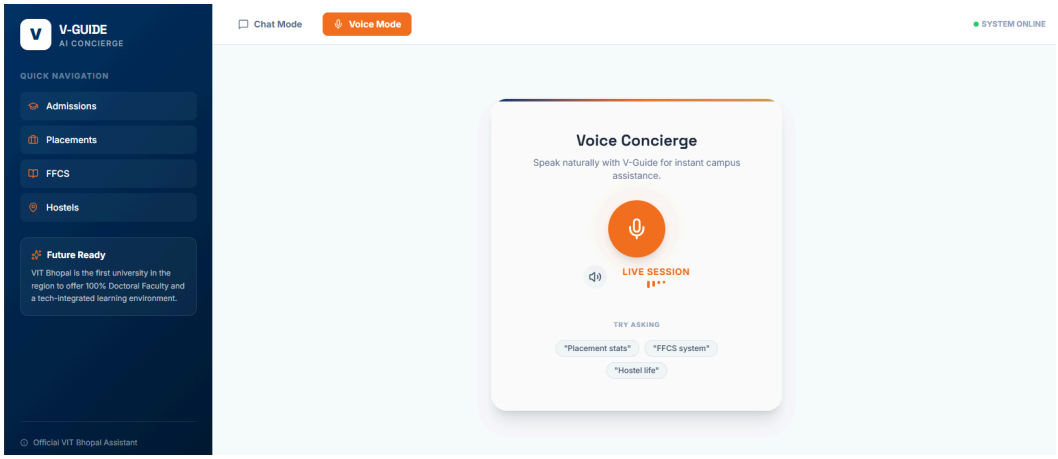
4. Website Images







5.Technical Architecture

Component	Technology
Frontend	React.js / Tailwind CSS
AI Logic	LLM (GPT-4o / Gemini 1.5 Pro)
Voice Engine	Web Speech API / ElevenLabs
Knowledge Base	Vector DB (Pinecone) + Web Scraper

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6. Directory Structure

Plaintext

```
VIT-Bhopal-AI-Concierge/
├── components/        # Chat and Voice UI Logic
├── data/              # Indexed university transcripts
├── system_prompt.md   # AI Personality & Guardrails
└── styles/            # UI Positioning & Animations
```

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7. Conclusion

The **VIT-Bhopal-AI-Concierge** successfully streamlines information access for the VIT Bhopal community. By combining a professional "Future Ready" personality with robust RAG technology, it provides a reliable and engaging user experience.